

Feature Engineering Report

TASK 1: FEATURE CATEGORIZATION

Classified into 15 Numerical, 6 Categorical, 3 Binary features.

TASK 2: BMI FEATURE ENGINEERING

Created 'bmi_category' and 'bmi_risk_score'. Medical Reasoning: BMI groups reflect non-linear metabolic syndrome thresholds. Obesity is a step-function risk multiplier.

TASK 3: AGE FEATURE ENGINEERING

Created 'age_group' and 'age_risk_score'. Captures the compounded biological deterioration inherent to aging.

TASK 4: CARDIOVASCULAR RISK INDEX

Created 'cardiovascular_risk_index' (0-100). Composite scoring of lipids, blood pressure, and smoking.

TASK 5: METABOLIC RISK INDEX

Created 'metabolic_risk_index' (0-100). Highlights systemic insulin resistance.

TASK 6: LIFESTYLE RISK INDEX

Created 'lifestyle_risk_index' mapping detrimental habits to risk scale.

TASK 7: OVERALL HEALTH SCORE

Created 'health_score' (0-100) and 'health_category'. Inverse metric derived from all composite risks.

TASK 8: INTERACTION FEATURES

Added pairwise interactions (e.g., bmi_glucose_interaction) to capture synergistic effects between variables before applying non-linear models.

TASK 9: ENCODING

Applied One-Hot Encoding to categorical variables. Label encoding was evaluated but OHE preferred for non-ordinal medical factors.

TASK 10: FEATURE SCALING

Compared standard, min-max, and robust scalers. Applied RobustScaler across numerals to mitigate impact of extreme pathological values.

TASK 11: FEATURE SELECTION

Evaluated features using RF Importance. Top 10 predictors for diabetes risk:

fasting_glucose, bmi_glucose_interaction, age_bmi_interaction, glucose_hba1c_interaction,
metabolic_risk_index, age_bp_interaction, age_cholesterol_interaction, age, health_score, weight_kg.